



Aryash Srivastava
Energy Science and Engineering
Indian Institute of Technology Bombay

22B1506
B.Tech.
Gender: Male
DOB: 06/05/2004

Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2026	
Intermediate	CBSE	Alard Public School	2022	
Matriculation	CBSE	Podar International School	2020	

Pursuing a Minor Degree in **Artificial Intelligence & Data Science** under the department of C-MInDS, IIT Bombay

SCHOLASTIC ACHIEVEMENTS

- Obtained **99.77 percentile** in JEE Mains 2022 Exam competing with over 1 million candidates (2022)
- Ranked in **top 1.4%** of JEE Advanced 2022 Exam, competing with over 0.15 million candidates (2022)
- Secured **9.4+** SPI in 4th semester | Grabbed **AA/AB** grade in **26** courses for academic excellence (2025)

PROFESSIONAL EXPERIENCE

Simulation Software Engineer | Under Research & Development Department

(Jun-Jul 2025)

Nikko Co. Ltd. | Hyogo, Japan

- Simulated a **G-LiNB burner** on the concept of **Lagrangian Multiphase Model** using **StarCCM+**, after building it on Autodesk Inventory, ensuring stability under high speed fluid dynamics & improved **devolatilization rates**
- Innovated the '**Swivel Feather**' blade design which was **approved for deployment in operational systems** & was responsible for providing (swirl) better aerodynamic powder dispersion performing better in Flame Stability
- Worked in team of 5, iterating Parcel Streams, Growth Rate & Prism Layer to enhance **Thermal Gravity Results**

Data Analyst Intern

(May-Jun 2024)

NoQs Digital | Awarded a **Letter of Recommendation & Stand out Performer Award**.

- Assisted Product Managers to oversee thousands of tasks by developing **Dynamic Dashboard** on Google Sheets
- Solved a Case Study for the status of fixed deposits for an individual, focusing on linked FDs and 15G FY forms
- Created **PowerBI** Dashboard to predict best banks profits by period and display metrics like matured policies

KEY PROJECTS

Feeble Audio Transcript Generator | IE-643 Deep Learning - Theory & Practice

(Sep 2024-Dec 2024)

Course Project | Instructor: Prof. Balamurugan

- Developed an algorithm to transcribe distorted audio, Generated **10+** hrs of data by augmenting **Mozilla Common Voice** dataset by addition or replacement by gaussian white noise for training a generalized model
- Detected noisy region via **spectrogram decomposition** and reconstructed missing words by **fill-mask** logic

Summary Categorizer | DS-203 Programming for Data Science

(Mar 2025-May 2025)

Course Project | Instructor: Prof. Vinay Kulkarni

- Preprocessing **1K+** summaries, tested 3+ embeddings (BoW, TF-IDF, Word2Vec), built a combined **hybrid model**
- Used UMAP to create **Word Clouds** per session and implemented ranking logic for **top 3** summaries per keyword
- Developed a **Python app** for keyword search & visualization, validated with **Silhouette Score (0.27)**

Image Caption Generator | Seasons of Code 2024

(May 2024-Jul 2024)

Web n Coding Club

- Built an architecture integrating **VGG16** for image feature extraction and **LSTM network** for text generation
- Used the Flickr8k dataset to train the model on a wide spectrum of **visual scenarios and linguistic expressions**

Compress Me If You Can | CS-419 Intro to Machine Learning

(Mar 2024- May 2024)

Course Project | Guide: Prof. Abir De, Department of Computer Science & Engineering

- Tried methods such as **Discrete Cosine Transform** taken by Huffman Coding & **AutoEncoders** to compress data.
- Chose **AutoEncoders** over Huffman Coding to save computations & Obtained **95% reduction in memory**
- Generated a new image dataset from the MNIST dataset using **Generative Adversarial Networks (GANs)**

Real time Face Recognition System | Winter in Data Science 2023

(Dec 2023-Jan-2024)

Analytics Club

- Used Siamese Network to train **One shot Learning** Model for generating Real-Time Face Recognition System
- Employed Transfer Learning by importing **VGG16** inbuilt model & enhancing the accuracy from **68% to 88%**
- Performed Real-Time Test of the model using **OpenCV** & Hyper-parameter tuning for enhancing Precision

Probability Theory : Learning Project | Summer Of Science 2024

(May 2024-Jul 2024)

Maths & Physics Club

- Covered **Bayesian Models** for prior/posterior distributions & Gaussian Kernels in Increasing the Feature Space.
- Researched upon Risk Minimization & gained depth knowledge on Convergence & **Fundamental Theory of PAC**

COMPETITIONS.....

India Game Developer Conference (IGDC) | Inter-IIT Tech 2025

(Dec 2024-Jan 2025)

Team Bombay76

- Built a 2D Metroidvania-style game in Unity as part of a 6-member team under a defined problem statement
- Was responsible for each character's **Animation Controller Design & Cutscenes** for start/end of the game
- Built one complete game section, including platform layout, enemy placement and game logic **C#** scripting
- Implemented enemy **AI behaviors** for multiple enemy types, aligning them with level design mechanics

POSITION OF RESPONSIBILITY.....

Team ANYmotion | Member Of Theory Subsystem

(Feb 2024-Feb 2025)

- Built scientifically accurate simulations and animations using Python (**Manim & Blender** scripting)
- Built a project on **exoplanet transit light curve simulation**, real-time intensity graphs & brightness variation
- Scripted 3D physics-based (**2-body & Wrecking-Brick wall simulations**) to enhance Python automation skills

ACADEMIC PROJECTS.....

Cavitation in Centrifugal Pumps | EN 222 Fluids Mechanics & Heat Transfer

(Apr 2024-Jun 2024)

Course Project | Guide: Prof. Manaswita Bose, Department of Energy Science & Engineering

- Explored **bubble dynamics** to understand cavitation, including microjet formation generating shock waves
- Tuned different parameter & keep track of Total Head, Vapour Pressure to avoid cavitation in Centrifugal Pump
- Derived the **NPSHA** using Bernoulli & Darcy Weisbach Equations & the limitation in deriving **NPSHR**
- Compared the relation between NPSHA and various primary parameters such as Temperature, Vapour Pressure

Santa Fe City PV Modeling & Yield | EN790 Solar Photovoltaics: Photons to Farms

(Jan 2025-Apr 2025)

Course Project | Guide: Prof. Pradeep Nair, Department of Electrical Engineering

- Developed single-junction and tandem (Si-Perovskite) solar cells, incorporating recombination non-idealities and diode characteristics to compute max efficiency limits (**Shockley-Queisser Limits**)
- Implemented **NGSpice circuit** for series-connected panels, analyzing shadow-induced mismatch losses
- Validated shadow-mitigation (bypass diode configurations), improving panel efficiency in non-uniform irradiance
- Performed data-driven annual energy yield analysis on USA Santa Fe city using **NSRDB** irradiance datasets and **King's thermal model**, integrating temperature coefficient of efficiency (**TPCE**) for realistic field performance

OSeMOSYS net zero carbon simulation by 2050 | EN 401 Energy Systems Modelling

(May 2023)

Course Project | Guide: Prof. Pavan Hari, Department of Energy Science & Engineering

- Modeled World's energy system for 100% renewables by 2050 using OSeMOSYS with region, heat, and transport
- Processed multi-source datasets to fill gaps in annual demand, transport, and fuel parameters for better simulation
- Simulated electricity, heat, and transport — projecting 62.5% solar PV share in electricity and complete shift to electric heating by 2050
- Analyzed GHG reduction pathways, compared results with GENeSYS-MOD global projections

2-MW PV Farm Simulation | EN 318 Energy Systems Design Lab

(Jan 2025-Apr 2025)

Course Project | Instructor: Prof. Pavan Hari, Department of Energy Science & Engineering

- Modeled 2-MW PV system with PV arrays, Boost Converter (**MPPT Pando**), NPC inverter, and **25-kV** distribution
- Performed energy flow and fault analysis, expanding system capacity to **3 MW** analysing active/reactive power
- Designed **PI-based Vdc & Q controllers**, MPPT duty ratio, regulation, & **inverter modulation index**.
- Simulated in MATLAB/Simulink, analyzing DC-link stability, inverter dynamics, and grid fault transients

TECHNICAL SKILLS.....

- **Programming Languages:** C++ | Python | HTML | CSS | SQL
- **Libraries:** Numpy | Matplotlib | Pandas | scipy | Tensorflow | Manim | OpenCV | Pillow
- **Softwares & Packages:** Fusion 360 | $\LaTeX$ | Fractory | Blender | PowerBI | Autodesk Inventory | StarCCM+

CERTIFICATIONS.....

- Data Science for Python (*Learner Space IIT Bombay 2023*)
- Data Structures & Algorithms (*Udemy*)
- Front-End Web Development using HTML, CSS & Java (*Johns Hopkins University*)

EXTRA-CURRICULARS.....

Tennis	• Secured 3rd place in Sports General Championship (Inter-Hostel Competition) Tennis • Reached Quarterfinals in Racketlon Sports Competition, playing Tennis & Badminton • Completed a 1 year long session National Sports Organisation in Lawn Tennis	(Feb 2024) (Apr 2023) (2022)
Culturals	• Participated in Gyrations an Inter-Hostel Dance Competition in General Championship • Performed in Annual Insync Dance Show for Fresher's Performance at the Institute • Recognized for vocal excellence with 3rd place in a Singing Competition conducted in school	(Oct-2024) (Apr 2022) (2017)
Others	• Built a start-up model from scratch for Startup & Pitching Competition, EnBuzz ECell • Constructed remote-controlled vehicle in XLR8 conducted by E&R Club • Performed Equity Research Analysis on PFC, event conducted by Finance Club	(Nov-2022) (Jan 2023) (Aug 2023)